

JEE MAIN REVISION MASTER: ELLIPSE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Ellipse: Eccentricity, Foci & Standard Properties	4 PYQs	Conics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ellipse: Eccentricity, Foci & Standard Properties	$e = \sqrt{1 - \frac{b^2}{a^2}}, \quad \text{Foci: } (\pm ae, 0), \quad \text{Latus Rectum} = \frac{2b^2}{a}, \quad SP + S'P = 2a$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ellipse: Eccentricity, Foci & Standard Properties (Moderate Yield — 4 PYQs)

Core Principle: Standard ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$): Eccentricity $e = \sqrt{1 - \frac{b^2}{a^2}} < 1$. Foci $(\pm ae, 0)$, Directrices $x = \pm a/e$, Length of major axis = $2a$, minor axis = $2b$, Length of latus rectum = $(2b^2/a)$. Sum of focal distances of any point is constant: $SP + S'P = 2a$.

■ **Caution / JEE Pitfall:** If $b > a$ (vertical ellipse), major axis is along y-axis: $e = \sqrt{1 - a^2/b^2}$ and foci are $(0, \pm be)$!

■ **Speed Trick:** Distance between foci = $2ae$; Distance between directrices = $2a/e$.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "The length of the latus rectum of $x^2+3y^2=12$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If the area of the ellipse is $\frac{\pi}{225} + \frac{\pi}{25}\lambda^2 = 1$ is 20π sq units, then λ is..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "The eccentricity of the ellipse $9x^2+25y^2=225$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

The length of the latus rectum of $x^2+3y^2=12$ is

- (A) $(2/3)$ units
- (B) $(1/3)$ units
- (C) $\frac{4\sqrt{3}}{3}$ units
- (D) 24 units

Correct Answer: None

Solution Analysis: $\frac{x^2}{12} + \frac{y^2}{4} = 1$ L.R. = $\frac{2b^2}{a} = \frac{2 \times 4}{\sqrt{12}} = \frac{4\sqrt{3}}{3}$